

Отчёт к лабораторной работе 4

Тема – «Маскировка и анонимизация данных»

У меня возникли трудности с установкой расширения напрямую, поэтому я воспользовался docker:

- 1) Создал дамп базы данных:

```
pg_dump -U postgres -h localhost -p 5432 -d postgres > db.sql
```

- 2) Запустил docker с монтированием дампа:

```
aroslavsidakin@MacBookPro ~ % docker run -d --name pg-anonymizer \
-e POSTGRES_PASSWORD=mysecretpassword \
-p 6543:5432 \
-v /Users/aroslavsidakin/db.sql:/docker-entrypoint-initdb.d/init.sql \
registry.gitlab.com/dalibo/postgresql_anonymizer:latest
```

- 3) Подключился к базе данных и активировал Anonymizer:

```
docker exec -it pg-anonymizer psql -U postgres
postgres=# CREATE EXTENSION IF NOT EXISTS anon CASCADE;
[SELECT anon.init();
```

Динамическая маскировка данных:

```
postgres=# SECURITY LABEL FOR anon ON COLUMN users.surname
[IS 'MASKED WITH FUNCTION anon.partial(surname, 1, '*****', 0)'];
```

```
postgres=# SECURITY LABEL FOR anon ON COLUMN users.phone_number
[IS 'MASKED WITH FUNCTION anon.partial(phone_number, 5, '*****', 0)'];
```

```
postgres=# SECURITY LABEL FOR anon ON COLUMN users.email_address
[IS 'MASKED WITH FUNCTION anon.partial_email(email_address)'];
```

Проверка:

user_id	surname	name	patronymic	email_address	phone_number
1	O*****	Сергей	Иванович	***@mail.ru	+7952*****
2	D*****	Johnny		***@google.com	+7777*****
3	M*****	Александр	Владимирович	***@itmo.ru	+7812*****
4	C*****	Ярослав	Андреевич	***@yahoo.com	+7917*****
5	C*****	Иван	Петрович	***@mail.ru	+7800*****

Materialized Views:

1) Generalization:

```
postgres=# CREATE MATERIALIZED VIEW mv_flight_quarters AS
SELECT
    flight_id,
    number,
    tstzrange(
        date_trunc('quarter', departure_date),
        date_trunc('quarter', departure_date) + interval '3 months',
        '[')
    ) AS departure_quarter,
    tstzrange(
        date_trunc('quarter', arrival_date),
        date_trunc('quarter', arrival_date) + interval '3 months',
        '[')
    ) AS arrival_quarter,
    departure_airport_id,
    arrival_airport_id,
    airplane_id
FROM flight;
```

```
postgres=# SELECT * FROM mv_flight_quarters;
```

flight_id	number	departure_quarter	arrival_quarter	departure_airport_id	arrival_airport_id	airplane_id
1	N4-254	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	1	2	1
2	M7-250	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	3	5	4
3	M5-000	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	4	2	5
4	A7-004	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	5	1	1
5	N8-001	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	["2024-01-01 00:00:00+00", "2024-04-01 00:00:00+00")	4	3	3

(5 rows)

2) Randomization:

```
postgres=# CREATE MATERIALIZED VIEW anon_usersbooking_randomized AS
SELECT
    booking_id,
    user_id,
    CONCAT(
        chr(65 + floor(random() * 26)::int),
        (1 + floor(random() * 99))::text
    ) AS randomized_place_number
FROM usersbooking;
```

```
[postgres=# SELECT * FROM anon_usersbooking_randomized;
```

booking_id	user_id	randomized_place_number
1	1	C87
1	2	F22
5	3	I49
4	4	A97
2	5	Y10
3	5	Y65

(6 rows)

3) Generic Hashing:

```

postgres=# CREATE MATERIALIZED VIEW anon_flights_hashed AS
SELECT
  flight_id,
  encode(hmac(
    number::bytea,
    current_setting('anon.salt')::bytea,
    'sha256'
  ), 'hex') AS hashed_flight_number,
  departure_date,
  arrival_date,
  departure_airport_id,
  arrival_airport_id,
  airplane_id
[FROM flight;

```

```

postgres=# SELECT * FROM anon_flights_hashed;
 flight_id | hashed_flight_number | departure_date | arrival_date | departure_airport_id | arrival_airport_id | airplane_id
-----
1 | 75a0419c87314f544c65062959ab48d42e2194876a23b5d143a66be8dd6251b6 | 2024-03-19 14:55:00 | 2024-03-19 16:00:00 | 1 | 2 | 1
2 | 72ee457df99f9c741908993ead9f01a49d5cd01448f25a373016a015b95ba75ed | 2024-03-23 15:55:00 | 2024-03-23 17:35:00 | 3 | 5 | 4
3 | 827d90f87bc32087fc27f05104c5c0169ed2a588176e6b0c089e166e3b9899b18 | 2024-03-20 10:00:00 | 2024-03-20 11:20:00 | 4 | 2 | 5
4 | efdf2d573b17fabaaee41395a10b0267fe9e313e972cd75a435098e1b46b5d86f | 2024-03-19 12:25:00 | 2024-03-19 14:10:00 | 5 | 1 | 1
5 | 046423c942d9ae7457072c0d2ab9b70d92a00f22eb09ea5954bb1dc4b14427ac | 2024-03-24 13:40:00 | 2024-03-24 15:00:00 | 4 | 3 | 3
(5 rows)

```